

PYTHON LAB ASSIGNMENT EXPERIMENT-2

BY

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BATCH:- 1

1. Check whether the given number is divisible by 3 and 5 both.

→ Code:-

```
q1.py > _
1 #Raghav Vij 590023842
2 n = int(input("Enter number: "))
3
4 if n % 3 == 0 and n % 5 == 0:
5     print("Divisible by both 3 and 5")
6 else:
7     print("Not divisible by both")
8
```

→

→ Output:-

```
1.py
Enter number: 15
Divisible by both 3 and 5
```

→

2. Check whether a given number is multiple of five or not.

→ Code:-

```
q2.py > _
1 #Raghav Vij 590023842
2 n = int(input("Enter number: "))
3
4 if n % 5 == 0:
5     print("Multiple of 5")
6 else:
7     print("Not a multiple of 5")
8
```

→

→ Output:-

```
2.py
Enter number: 15
Multiple of 5
```

→

3. Find the greatest among the two numbers. If numbers are equal than print "numbers are equal".

→ Code:-

```
q3.py > _
1 #Raghav Vij 590023842
2 a = int(input("Enter first number: "))
3 b = int(input("Enter second number: "))
4
5 if a > b:
6     print("Greatest:", a)
7 elif b > a:
8     print("Greatest:", b)
9 else:
10    print("Numbers are equal")
11
```

→

→ Output:-

```
3.py
Enter first number: 10
Enter second number: 15
Greatest: 15
PS C:\Users\hp\Documents\UD
```

4. Find the greatest among three numbers assuming no two values are same.

→ Code:-

```
1 #Raghav Vij 590023842
2 a = int(input("Enter first number: "))
3 b = int(input("Enter second number: "))
4 c = int(input("Enter third number: "))
5
6 if a > b and a > c:
7     print("Greatest:", a)
8 elif b > a and b > c:
9     print("Greatest:", b)
10 else:
11     print("Greatest:", c)
12
```

→
→ Output:-

```
4.py
Enter first number: 10
Enter second number: 15
Enter third number: 20
Greatest: 20
PS C:\Users\hp\Document
```

5. Check whether the quadratic equation has real roots or imaginary roots. Display the roots.

→ Code:-

```
1 #Raghav Vij 590023842
2 import math
3
4 a = int(input("Enter a: "))
5 b = int(input("Enter b: "))
6 c = int(input("Enter c: "))
7
8 d = b*b - 4*a*c
9
10 if d > 0:
11     r1 = (-b + math.sqrt(d)) / (2*a)
12     r2 = (-b - math.sqrt(d)) / (2*a)
13     print("Real and different roots:", r1, r2)
14 elif d == 0:
15     r = -b / (2*a)
16     print("Real and same roots:", r)
17 else:
18     print("Imaginary roots")
19
```

→ Output:-

```
Enter a: 1
Enter b: 4
Enter c: 0
Real and different roots: 0.0 -4.0
```

6. Find whether a given year is a leap year or not.

→ Code:-

```
1 #Raghav V13 590023842
2 year = int(input("Enter year: "))
3
4 if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
5     print("Leap year")
6 else:
7     print("Not a leap year")
8
```

→

→ Output:-

```
Enter year: 2025
Not a leap year
```

→

7. Write a program which takes any date as input and display next date of the calendar

e.g.

I/P: day=20 month=9 year=2005

O/P: day=21 month=9 year 2005

→ Code:-

```
1 #Raghav V13 590023842
2 day = int(input("Enter day: "))
3 month = int(input("Enter month: "))
4 year = int(input("Enter year: "))
5
6 day = day + 1
7
8 if day > 30:
9     day = 1
10    month = month + 1
11
12    if month > 12:
13        month = 1
14        year = year + 1
15
16 print("Next date:", day, month, year)
17
```

→ Output:-

```
7.py
Enter day: 1
Enter month: 1
Enter year: 1
Next date: 2 1 1
```

→

PS C:\Users\hp\Docume

8. Print the grade sheet of a student for the given range of cgpa. Scan marks of five subjects and calculate the percentage.

CGPA=percentage/10

CGPA range:

0 to 3.4 -> F

3.5 to 5.0->C+

5.1 to 6->B

6.1 to 7-> B+

7.1 to 8-> A

8.1 to 9->A+

9.1 to 10-> O (Outstanding)

Sample Gradesheet

Name: Rohit Sharma

Roll Number: R17234512

SAPID: 50005673

Sem: 1

Course: B.Tech. CSE AI&ML

Subject name: Marks

PDS: 70

Python: 80

Chemistry: 90

English: 60

Physics: 50

Percentage: 70%

CGPA:7.0

Grade:

→ Code:-

```
1 #Raghav Vij 590023842
2 name = input("Enter name: ")
3 roll = input("Enter roll number: ")
4 sapid = input("Enter SAP ID: ")
5
6 m1 = int(input("Enter PDS marks: "))
7 m2 = int(input("Enter Python marks: "))
8 m3 = int(input("Enter Chemistry marks: "))
9 m4 = int(input("Enter English marks: "))
10 m5 = int(input("Enter Physics marks: "))
11
12 total = m1 + m2 + m3 + m4 + m5
13 percentage = total / 5
14 cgpa = percentage / 10
15
16 if cgpa <= 3.4:
17     grade = "F"
18 elif cgpa <= 5.0:
19     grade = "C+"
20 elif cgpa <= 6.0:
21     grade = "B"
22 elif cgpa <= 7.0:
23     grade = "B+"
24 elif cgpa <= 8.0:
25     grade = "A"
26 elif cgpa <= 9.0:
27     grade = "A+"
28 else:
29     grade = "O"
30
31 print("\nName:", name)
32 print("Roll Number:", roll, "SAPID:", sapid)
33 print("Percentage:", percentage, "%")
34 print("CGPA:", cgpa)
35 print("Grade:", grade)
36
```

→

→ Output:-

```
Enter name: Rohit Sharma
Enter roll number: R17234512
Enter SAP ID: 50005673
Enter PDS marks: 70
Enter Python marks: 80
Enter Chemistry marks: 90
Enter English marks: 60
Enter Physics marks: 50

Name: Rohit Sharma
Roll Number: R17234512 SAPID: 50005673
Percentage: 70.0 %
CGPA: 7.0
Grade: B+
```

→